

Hexadecimal Addition

$$\text{base}(r) = 16$$

$$0 \text{ to } (16-1)$$

$$0 \text{ to } 15$$

$$26 = \underline{1} \times 16 + \underline{10}$$

$$r = 26 - 16$$

$$r = 10$$

$$1 + 13 + 10 = 24$$

$$13 \times 16^0 + 13 \times 16^0$$

$$26 \times 16^0$$

$$(16 + 10) \times 16^0$$

$$16 \times 16^0 + 10 \times 16^0$$

$$1 \times 16^1 + 10 \times 16^0$$

carry

0
1
2
3
4
5
6
7
8
9
A
B
C
D
E
F

10
11
12
13
14
15

i)

$$\begin{array}{r} 5 6 8 9 \\ + 4 5 7 4 \\ \hline 9 B F D \\ \hline \end{array} \text{Ans}$$

$$\begin{array}{r} F) 30 \\ F) 10 \\ \hline + A) 10 \\ \hline 40 = \underline{2} \times 16 + 8 \end{array}$$

ii)

$$\begin{array}{r} 1 1 \\ 16^2 16^1 16^0 \\ \downarrow \downarrow \downarrow \\ 1 1 \\ A D D \\ + 1 \downarrow D A D \\ \hline 1 8 8 A \\ \hline \end{array} \text{Ans}$$

$$\begin{array}{r} F \\ + F \\ \hline 30 = \underline{1} \times 16 + 14 \end{array}$$

$24 = \underline{1} \times 16 + 8$
 $2 \times 16 = \underline{32}$
 $30 = \underline{1} \times 16 + 14$

4
5
6
7
8
9
A
B
C
D
E
F

10
11
12
13
14
15

$$\begin{array}{r} 5 \ 6 \ 8 \ 9 \\ + \ 4 \ 5 \ 7 \ 4 \\ \hline 9 \ B \ F \ D \\ \hline \end{array} \text{Ans}$$

$$\begin{array}{r} A \ D \ D \\ + \ D \ A \ D \\ \hline 1 \ 8 \ 8 \ A \\ \hline \end{array} \text{Ans}$$

$$\begin{array}{r} F \) \ 30 \\ F \) \ 10 \\ + \ A \) \ 10 \\ \hline 40 = 2 \times 16 + 8 \end{array}$$

$$\begin{array}{r} F \\ + \ F \\ \hline 30 = 1 \times 16 + 14 \end{array}$$

$24 = 1 \times 16 + 8$
 $2 \times 16 = 32$
 $30 = 1 \times 16 + 14 \text{ E}$

$x = 26 - 16$
 $x = 10$
 $L + 13 + 10 = 24$

$$13 \times 16^0 + 13 \times 16^0$$

$$26 \times 16^0$$

$$(16 + 10) \times 16^0$$

$$16 \times 16^0 + 10 \times 16^0$$

$$1 \times 16^1 + 10 \times 16^0$$

carry

How i)

$$\begin{array}{r} 8 \ 9 \ 9 \\ + \ 1 \ 8 \ 9 \\ \hline \end{array}$$

ii)

$$\begin{array}{r} D \ A \ F \\ + \ B \ A \ F \\ \hline \end{array}$$

Hexadecimal Subtraction

Show

$$19 - 7 = 12 \quad B < C$$

$$22 - 8 = 14$$

$$16 + 6 = 22 \quad 16 + 3 = 19 \quad 16 + 11 = 27$$

$$27 - 12 = 15$$

i)

$$\begin{array}{r} 9654 \\ - 5321 \\ \hline 4333 \end{array}$$

Ans

Q:-2

$$\begin{array}{r} FAB \\ - AEF \\ \hline \end{array}$$

Q:-3

$$\begin{array}{r} 7788 \\ - DE F \\ \hline \end{array}$$

ii)

$$\begin{array}{r} 896734B \\ - 587C \\ \hline 3ECF \end{array}$$

Ans

16×16^6

3×16^1

$B \times 16^0$

Q:-1

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
A	- 10
B	- 11
C	- 12
D	- 13
E	- 14
F	- 15

Show

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
A	- 10
B	- 11
C	- 12
D	- 13
E	- 14
F	- 15

i)

16^2 16^1 16^0
 $\downarrow$ $\downarrow$ $\downarrow$
 1
 9 4
 $\times$ 1 2

 1 2 8
 9 4 $\times$

 A 6 8

 Any

$$\frac{1 \times 16^1 \times 4 \times 16^0}{4 \times 16^1}$$

ii)

	¹	¹⁰	¹¹	¹²
	A	B	C	
x		2	3	
	2	0	3	4
	1	5	7	8
				x
	1	7	7	B
	1	1	1	1
	16 ⁴	16 ²	16 ²	16 ¹

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$$\begin{array}{r} 818 \\ \times 21 \\ \hline \end{array}$$

h)

DAF
x54

$$36 = 2 \times 16 + 4$$

$$35 = \underline{2} \times 16 + \underline{3}$$

$$32 = 2 \times 16 + 0$$

$$24 = 1 \times 16 + 8$$

$$23 = 1 \times 16 + 7$$

$$21 = 1 \times 16 + 5$$